

Studio 2 • Name the Pieces

OPIM 5641 • Business Decision Modeling
warm-up with your partner • pencil only • ~15 min

Names: _____

Flair's Furniture makes inexpensive **tables** and **chairs**. Each table takes **3 hours** of carpentry and **2 hours** of painting. Each chair takes **4 hours** of carpentry and **1 hour** of painting. This month there are **2,400 carpentry hours** and **1,000 painting hours** available. Marketing wants **no more than 450 chairs** (there is already a pile of them) and **at least 100 tables** (inventory is low). Each table earns **\$7** profit; each chair earns **\$5**.

1 • Decision variables what do we get to choose?

2 • Objective function what are we making as big as possible?

Max $Z =$ _____ T + _____ C

The two numbers you just wrote are the **objective function coefficients**. In one sentence, what does the 7 actually mean? _____

3 • Constraints what rules can we not break?

_____ T + _____ $C \leq$ _____ (_____)

_____ T + _____ $C \leq$ _____ (_____)

_____ $\leq$ _____ (_____)

_____ $\geq$ _____ (_____)

$T, C \geq$ _____ (_____)

4 • Try two plans is it legal? what does it pay?

A manager suggests **200 tables and 100 chairs**. Another suggests **500 tables and 450 chairs**. Check each against every constraint, then compute the profit of the legal one.

Next: open **Studio2_TwoWays_blank.ipynb**. We will find the best plan two ways — by brute force (let the computer try every plan) and graphically (draw it and check the corners) — and see that they agree.